Experimental Skills and Data Analysis

International Physics Olympiad · Lecture + Problems

Introductory study guide · 6 original problems

> Organized around the IPhO syllabus. This guide is not exhaustive; consult the syllabus study PDF and the IPhO syllabus webpage for the full scope and qualifications.

3.2 Safety

Identify hazards before measuring; use safe voltages, avoid viewing bright sources directly, and follow equipment instructions. Do not improvise with hazardous apparatus.

3.3 Measurement techniques and apparatus

Choose suitable instruments and ranges, account for zero offsets and instrument loading, and record units. Plan repeated measurements and a control where practical.

3.4 Accuracy

Measure many periods rather than one to reduce timing uncertainty. Sample enough points to resolve the behaviour, especially near rapid changes; report sensible significant figures.

3.5 Experimental uncertainty analysis

Separate random scatter from systematic bias. Repetition can reduce random uncertainty in the mean but does not eliminate a common calibration error. Propagate uncertainties using a stated method and identify the dominant contribution.

3.6 Data analysis

Linearize a model when useful. Plot measured points and error bars with labeled axes and units; fit a line, interpret its slope and intercept, and estimate uncertainty using a reasonable graphical or statistical method.

Problems

Try each problem before opening the worked solutions. Exercises are original MAIS PAC teaching material, not official IPhO problems.

1. Twenty oscillations take 40.0 s. Find the period.

2. A length is (10.0 ± 0.1) cm. What is its relative uncertainty?

3. For $y = ax + b$, two points on a fitted line are (1, 3) and (3, 7). Find slope and intercept.

4. Every reading has the same unknown zero offset. Will repeating the measurement remove it?

5. Challenge 1 — A pendulum has $T^2 = (4\pi^2/g)L$. A line fit of T^2 versus L gives slope (4.00 ± 0.08) s^2/m. Estimate g and its uncertainty using first-order propagation.

6. Challenge 2 — A resistor is measured with a voltmeter of finite resistance connected in parallel with it, and an ammeter measuring the total current supplied to both branches. Does V/I overestimate or underestimate the resistor’s true resistance? Explain.